

Sync or Sink

Labor Market

Analysis

Graduation Project · Business Intelligence & Data Analytics

Team: Sync or Sink

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The Problem We're Solving

Labor markets are shifting faster than ever. Job seekers, employers, and policymakers lack the real-time, data-driven clarity needed to make informed decisions — leading to mismatches, inefficiencies, and missed opportunities.

Information Asymmetry

Job seekers and employers operate without shared, reliable labor market intelligence.

Lack of Predictive Insight

Most available data is historical — not forward-looking or actionable.

Fragmented Data Sources

Labor data is scattered across platforms, formats, and agencies with no unified view.



Project Objectives

Six strategic goals driving this end-to-end analytics initiative.



Consolidate Data

Unify fragmented labor datasets into a single, reliable source.



Clean & Standardize

Transform raw data into analysis-ready format.



Build Interactive Dashboards

Develop Power BI dashboards for real-time exploration.



Identify Market Trends

Surface hiring patterns, wage trends, and demand signals.



Drive Actionable Recommendations

Deliver guidance for job seekers and employers.

Dataset Overview

	A	B	C	D	E	F	G	H	I	J	K	L
	job_title	experience_years	education_level	skills_count	industry	company_size	location	remote_work	certification	salary	Experience_Level	Skill_Level
1	AI Engineer	10	Bachelor	2	Healthcare	Medium	India	Hybrid	2	109413	Lead	Beginner
2	Data Analyst	5	Bachelor	17	Telecom	Small	Australia	No	0	93764	Senior	Expert
3	Frontend Developer	18	PhD	4	Media	Medium	Singapore	No	1	148123	Lead	Intermediate
4	Business Analyst	19	PhD	13	Retail	Medium	Canada	Yes	0	189123	Lead	Expert
5	Product Manager	15	Bachelor	7	Manufacturing	Large	Sweden	Yes	0	165069	Lead	Advanced
6	AI Engineer	0	High School	4	Education	Enterprise	USA	No	2	180351	Junior	Intermediate
7	Backend Developer	6	High School	16	Manufacturing	Enterprise	Canada	No	3	165375	Senior	Expert
8	Machine Learning Engineer	4	Bachelor	18	Healthcare	Enterprise	Canada	Hybrid	5	202463	Mid-Level	Expert
9	AI Engineer	5	Diploma	14	Telecom	Enterprise	Singapore	Yes	0	171834	Senior	Expert
10	Frontend Developer	18	PhD	2	Retail	Enterprise	India	Hybrid	5	128377	Lead	Beginner
11	Frontend Developer	1	High School	10	Healthcare	Large	Netherlands	Yes	5	105755	Junior	Advanced
12	Master	19	Master	7	Finance	Large	Netherlands	Yes	5	157417	Lead	Advanced
13	Product Manager	8	High School	3	Healthcare	Medium	Canada	Yes	3	154190	Lead	Beginner
14	Data Analyst	3	Bachelor	3	Education	Startup	Singapore	Yes	4	97600	Mid-Level	Beginner
15	Business Analyst	14	PhD	5	Healthcare	Enterprise	India	No	1	105481	Lead	Intermediate
16	Business Analyst	0	Bachelor	9	Technology	Medium	Remote	No	4	85143	Junior	Advanced
17	Machine Learning Engineer	11	PhD	16	Consulting	Medium	Germany	Hybrid	3	191757	Lead	Expert
18	DevOps Engineer	10	Bachelor	1	Education	Large	UK	No	2	155931	Lead	Beginner
19	Product Manager	9	Master	12	Finance	Small	Singapore	Yes	1	141207	Lead	Expert
20	Software Engineer	11	High School	10	Media	Large	Sweden	No	1	124532	Lead	Advanced
21	Business Analyst	4	Bachelor	15	Retail	Medium	UK	Hybrid	0	108203	Mid-Level	Expert
22	Cybersecurity Analyst	17	Diploma	7	Government	Small	Germany	Hybrid	5	161268	Lead	Advanced
23	DevOps Engineer	10	Bachelor	8	Government	Small	Germany	Yes	1	151828	Lead	Advanced
24	Software Engineer	5	High School	2	Retail	Enterprise	Sweden	Hybrid	1	115827	Senior	Beginner
25	Cybersecurity Analyst	12	High School	15	Technology	Small	Australia	Hybrid	1	125030	Lead	Expert
26	Product Manager	13	Master	8	Retail	Medium	India	Hybrid	3	113838	Lead	Advanced
27	Data Scientist	2	Bachelor	14	Education	Startup	Remote	Yes	5	109478	Mid-Level	Expert
28	Cybersecurity Analyst	9	PhD	19	Technology	Medium	Germany	No	2	169253	Lead	Expert

Source & Scope

Kaggle-Job salary prediction dataset containing 250,000 job records across multiple industries, locations, education levels, and experience levels. The dataset was cleaned and prepared for analytics and machine learning. — covering 2021–2026.

250K+
Job Records

10
Industry Sectors

10
Locations

5
Years Covered

Data Cleaning & Preprocessing

A rigorous three-stage pipeline transformed raw, inconsistent data into a trusted analytical foundation.



Each stage included quality gates — row counts validated, schema integrity confirmed, and null rates tracked — ensuring only clean, reliable data advanced to modeling.

```
File Edit Selection View Go Run Terminal Help
preprocessing.ipynb
E > Sync or Slink > python > preprocessing.ipynb > import pandas as pd
Generate + Code + Markdown | Run All Clear All Outputs | Outline ...

import pandas as pd

df = pd.read_csv("job_salary_prediction_dataset.csv")

df.head()

df.tail()

df.shape

df.info()
```

	job_title	experience_years	education_level	skills_count	industry	company_size	location	remote_work	certifications	salary
0	AI Engineer	10	Bachelor	2	Healthcare	Medium	India	Hybrid	2	109413
1	Data Analyst	5	Bachelor	17	Telecom	Small	Australia	No	0	93764
2	Frontend Developer	18	PhD	4	Media	Medium	Singapore	No	1	148123
3	Business Analyst	19	PhD	13	Retail	Medium	Canada	Yes	0	189123
4	Product Manager	15	Bachelor	7	Manufacturing	Large	Sweden	Yes	0	165069

	job_title	experience_years	education_level	skills_count	industry	company_size	location	remote_work	certifications	salary
249995	Software Engineer	17	PhD	2	Telecom	Enterprise	India	No	1	127791
249996	Frontend Developer	20	PhD	7	Telecom	Startup	Remote	No	2	154593
249997	Business Analyst	1	Bachelor	12	Retail	Enterprise	India	Yes	0	75988
249998	Data Scientist	0	High School	2	Consulting	Small	Sweden	Hybrid	5	90467
249999	Data Analyst	16	Diploma	2	Technology	Medium	UK	No	5	133084

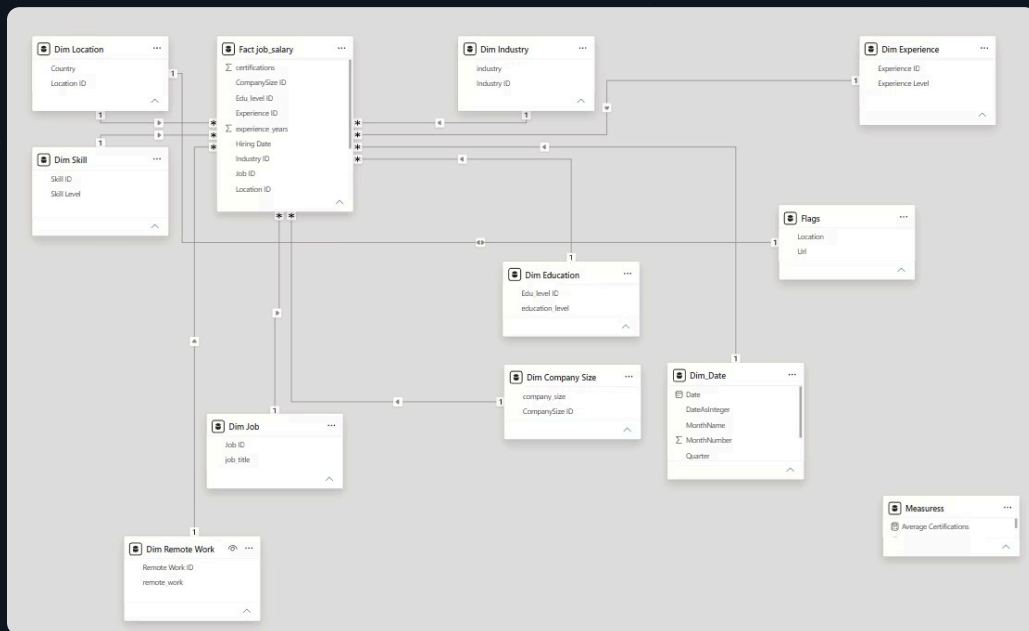

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 250000 entries, 0 to 249999
Data columns (total 10 columns):
 #   Column      Non-Null Count  Dtype
---  ---
 0   job_title   250000 non-null object
```

Data Cleaning Steps

Loaded the dataset using Pandas.

- Explored the dataset structure (info(), shape, describe()).
- Checked for missing values.
- Removed duplicate records.
- Standardized text columns by trimming extra spaces.
- Verified data types and unique values before analysis.

Data Modeling Architecture



Star Schema Design

A centralized fact table connects to four dimension tables — enabling fast, flexible analysis across time, geography, industry, and role type.

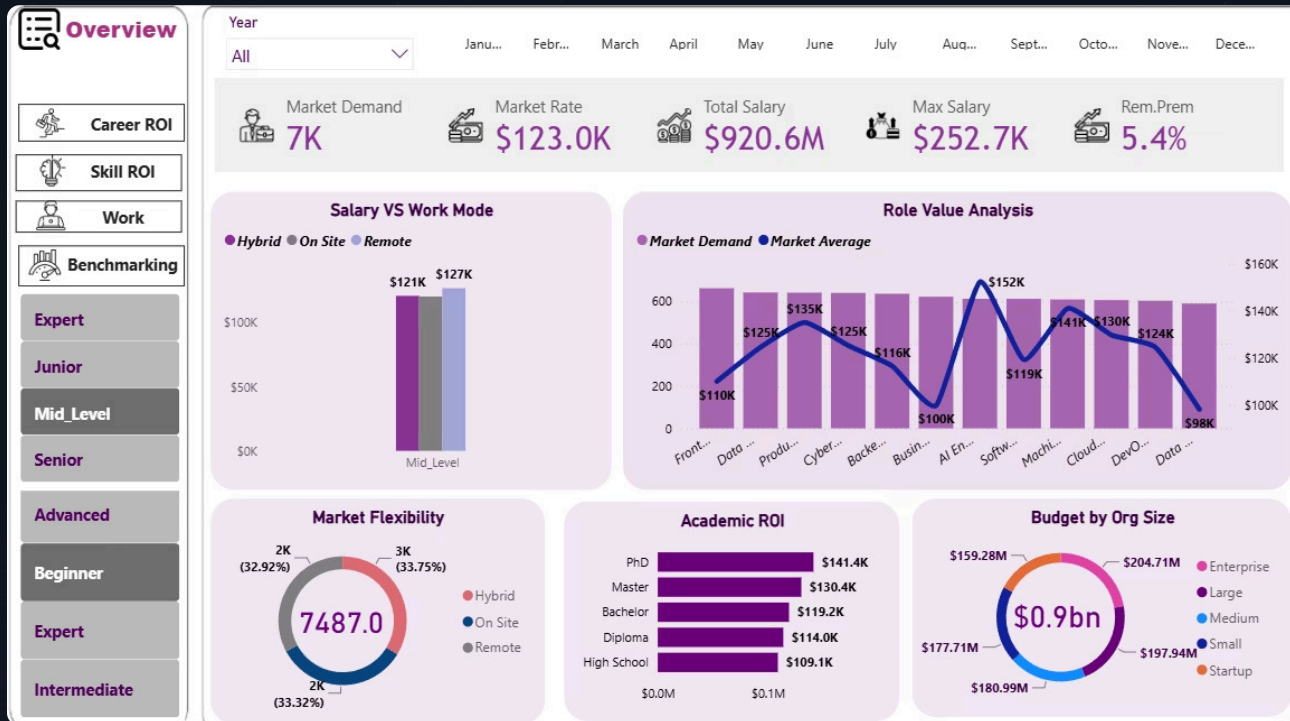
DAX Measures

Custom calculations for YoY growth, wage indices, and demand ratios.

Power Query

Automated refresh pipelines with parameterized data loading.

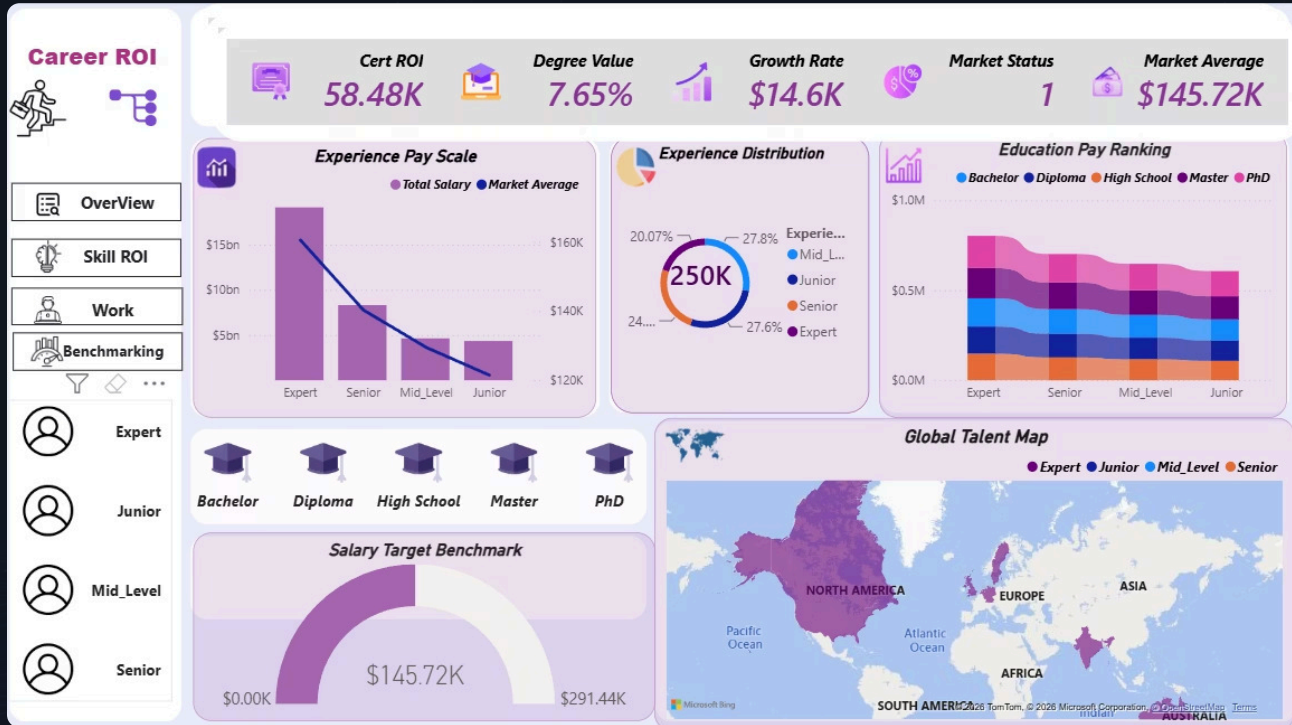
Dashboard 1 — Labor Market Overview



Key Insights

- Overall market demand reached **7K job opportunities**.
- Total salary value exceeded **\$920M**, with a maximum salary of **\$252.7K**.
- Hybrid and Remote** work modes offer competitive salaries compared to On-site roles.
- Higher education levels (Master's and PhD) are associated with higher salary ranges.
- Large and Enterprise companies account for the largest share of the salary budget.

Dashboard 2 — Career ROI



Key Insights

- Median salary gap between top and The market average salary is approximately \$145.7K.
- Professionals with Expert-level experience show the highest total salary values.
- Higher education levels, especially Master's and PhD, are associated with stronger salary outcomes.
- The dashboard tracks talent demand across multiple global regions through the Global Talent Map.
- The current certification ROI is estimated at \$58.5K, indicating a positive return from skill development.

Dashboards 3 — Skills ROI



Key Insights

- The average salary reached \$159.3K across all skill levels.
- Professionals possess an average of 10 skills, highlighting the importance of diversified skill sets.
- Expert-level professionals earn the highest average salaries compared to other skill levels.
- 723 respondents were included in the skills analysis, providing a reliable overview of workforce capabilities.

Dashboards 4 — Benchmarking



Key Insights

- Python, SQL, and Cloud appear in 67%
- The selected role has 20,945 employees with a balanced market status.
- Average salary is \$173K, ranging from \$66K to \$333K.
- Remote, Hybrid, and On-site work are distributed almost evenly across employees.
- Employee distribution spans multiple countries, reflecting global demand.

Business Insights & Recommendations

Salary Growth

- Expert professionals earn approximately **35% higher salaries** than Junior professionals.
- The average market salary reached **\$159.3K** across analyzed roles

Education Impact

- Master's and PhD holders consistently achieve the highest salary levels.
- Professional certifications contribute an estimated **\$58.48K** in career value..

Skills & Experience

- Professionals possess an average of **10 skills**.
- Higher skill levels are strongly associated with increased salaries

Workforce Trends

- Hybrid and Remote work models remain competitive with On-site positions.
- Talent demand is distributed across multiple countries and industries..

For Job Seekers

- Focus on developing advanced technical skills.
- Earn industry-recognized certifications.
- Build experience through projects and internships.

For Companies

- Invest in employee upskilling.
- Offer flexible work arrangements.
- Use data-driven insights for recruitment and salary planning.

Thank You

We appreciate your time and attention. We welcome your questions, feedback, and discussion.

QUESTIONS?

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